

How to evaluate the speed of a bike

QuBi

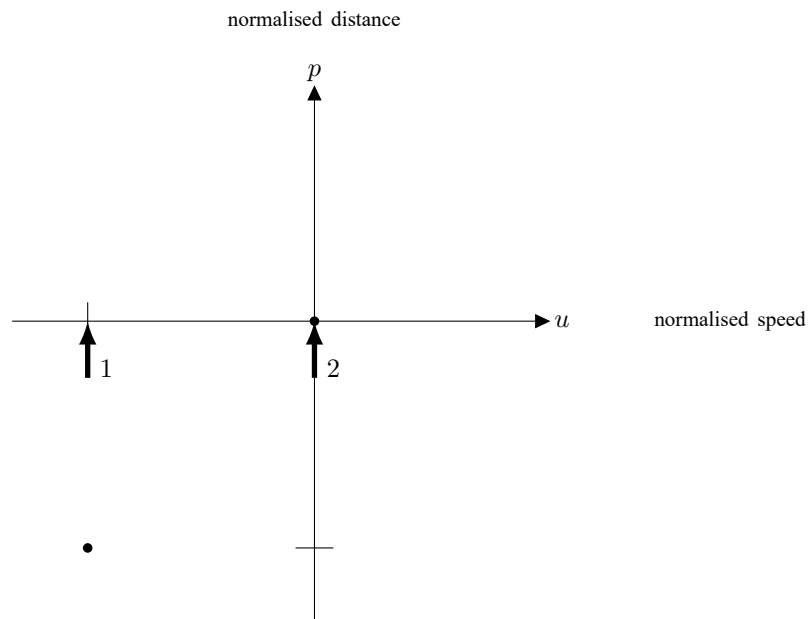
June 25, 2026

The usual tachometers you can find for your bike are sluggish. They take forever to display the instantaneous speed. Probably because of some averaging process.

There must be a smarter way of approximating the speed. In 2026, we can certainly do a better refresh rate than less than 1 fps.

Description of the problem

In this article we describe an attempt at finding a "good" way of estimating the speed based on the common bike speed-o-meter setup: a magnetic sensor ticking on every spin of the wheel.



Therefore the kernel function must satisfy:

- $p(-1) = -1$

- $p(0) = 0$

But that doesn't yield any prediction capacity and is too permissive.

Let's get more precise. The dynamic resistance (so the gain in the I/V curve) R_C can be expressed explicitly as a function of I_C therefore giving a more refined description of the transfer curve.

For a BJT:

$$\begin{aligned}
 a &= \left. \frac{\partial I_C}{\partial V_{BE}} \right|_{V_{BE}=V_{BE}^0} \\
 &= \left. \frac{\partial}{\partial V_{BE}} \right|_{V_{BE}=V_{BE}^0} I_S \exp() \\
 &= \left(1 - \frac{2R_C}{2R_C + Z} \right) \Delta I \\
 &= \frac{Z}{2R_C + Z} \Delta I
 \end{aligned} \tag{1}$$

Limits

Large signal analysis

Unlike the Emitter Follower, there's no feedback here to rely on. Hence, we expect deviation from the model if the input is too wide.

Transistor spec mismatch

This section is TODO.

Example 12: the diode Ladder Filter

Description

As a follow-up of the Moog Ladder Filter, we now consider the case of the diode version. It looks very similar to the Moog one, except the BJTs have been replaced with diodes.

Example 13: the Moog Voltage Controlled High-Pass filter (module 904B)

Description

Similarly, Moog came up with a design for a high-pass filter (also patented). The structure differs a bit from the low-pass, but it is equally as puzzling.

Here below is one stage of the high-pass filter: